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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 7

Title: Developing finance KPI dashboards

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Experiment No.: 7

Developing finance KPI dashboards

Problem Statement:

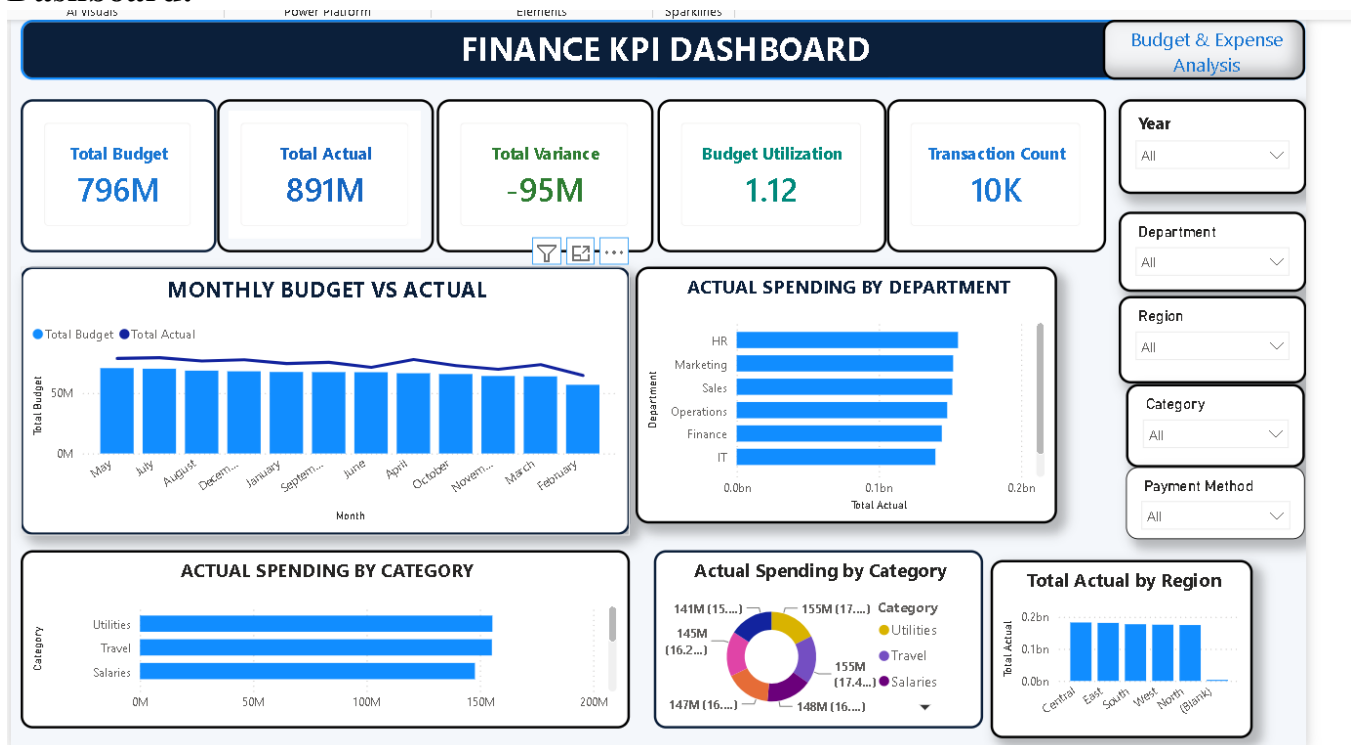
As a Financial Analyst in a manufacturing company, senior management needs clear visibility into how spending compares against approved budgets across departments, regions, and expense categories, since manually tracking thousands of transactions makes it difficult to spot overspending and support timely financial decisions. This project involves developing an interactive **Finance KPI Dashboard** in Power BI using the **Budget vs Actual Financial dataset** (Kaggle), tracking key metrics such as total budget, actual spend, budget gap, and budget utilization, while breaking down variance by department, region, category, and payment method to help management identify overspending and strengthen budget control.

Methodology / Steps Followed:

1. **Data Collection** – Obtained the Budget vs Actual Financial dataset from Kaggle, containing transaction-level records of budgeted and actual spending across departments, categories, regions, and payment methods (2021–2023).
2. **Data Cleaning & Preparation** – Imported the dataset into Power BI using Power Query, verified and corrected data types (Date, numeric amounts, text fields), and removed blank/null rows in Department, Category, and Region.
3. **Data Modeling** – Created a dedicated Date table using DAX (Year, Month Name, Quarter) and linked it to the Budget table via a one-to-many relationship to enable proper time-based filtering and sorting.
4. **Data Transformation** – Added calculated columns such as Variance, Variance %, and Budget Status (Over Budget / Under Budget) to support categorical and comparative analysis.
5. **DAX Measures Creation** – Built key measures including Total Budget, Total Actual, Total Variance, Budget Gap, Budget Utilization %, and Transactions to power the KPI cards and visuals.
6. **Dashboard Design** – Designed a two-page interactive dashboard (Overview and Budget Variance Analysis) incorporating KPI cards, donut charts, bar charts, and line charts to visualize budget performance, spend distribution, and trends over time.
7. **Interactivity** – Added slicers (Department, Category, Region, Payment Method) with cross-page slicer sync and a Reset Filters button, enabling dynamic filtering and exploration of the data.

8. **Insights & Analysis** – Analyzed the visualized data to extract key findings, such as departments and regions with the highest budget overrun, and payment methods driving actual spend.
9. **Version Control & Documentation** – Followed the GitHub workflow discussed in class: initialized a repository, structured project files (dataset, .pbix file, screenshots, README), committed changes, and pushed the final project to GitHub for submission.

Dashboard:



The Finance KPI Dashboard was developed in Power BI Desktop, built on the Budget vs Actual Financial dataset. The Overview page presents a complete financial performance snapshot, opening with five KPI cards — Total Budget (₹795M), Total Actual (₹890M), Budget Gap (-₹95M), Budget Utilization (112%), and Transactions (10K) — giving management an instant read on overall spending health.

Conclusion:

An interactive Finance KPI Dashboard was successfully developed in Power BI using the Budget vs Actual Financial dataset, enabling clear visibility into the organization's budget performance across departments, regions, categories, and payment methods. The Overview page consolidates key financial metrics — Total Budget, Total Actual, Budget Gap, Budget Utilization %, and Transactions — into a single, easy-to-navigate interface, supported by donut charts, trend lines, and bar charts that break down spending by category, department, region, and month.

The dashboard reveals that actual spending exceeds the approved budget overall, with certain departments and regions showing higher utilization and overspend than others, highlighting specific areas that require closer budget monitoring. Interactive slicers (Department, Category, Region, Payment Method) and a Reset button allow users to dynamically filter and explore the data, while the Key Insights panel summarizes the major findings for quick decision-making. Overall, the dashboard equips management with the visibility needed to track financial performance, identify overspending trends, and make informed, data-driven decisions to strengthen budget control and planning.

Learning Outcomes:

After completing this experiment, I learned:

- How to clean and transform transaction-level data, including handling data types and removing blank/null rows.
- How to build a Date table using DAX and create relationships for time-based analysis.
- How to create calculated columns for variance and budget status classification.
- How to write DAX measures for finance KPI calculations such as Total Budget, Total Actual, Budget Gap, and Budget Utilization %.
- How to design and format interactive KPI cards for a financial dashboard.
- How to create Donut, Bar, Column, and Line charts to visualize budget vs actual performance.
- How to analyze spending distribution across departments, regions, categories, and payment methods.
- How to identify and interpret budget variance and overspending trends.
- How to apply consistent formatting, color themes, and professional design principles to a Power BI dashboard.
- How to derive and present data-driven financial insights that support budget control and decision-making.

GitHub Repository: https://github.com/Balajivallepu/Data_Analytics-_for_Bussiness/blob/main/EXPERIMENT7/project_DAB_7.pbix

GitHub Repository Folder:

https://github.com/Balajivallepu/Data_Analytics-_for_Bussiness/tree/main/EXPERIMENT7